Q. 1(i)

In Exercises 1-2, suppose that A, B, C, D, and E are matrices with the following sizes:

A	B	C	D	E
(4×5)	(4×5)	(5×2)	(4×2)	(5×4)

In each part, determine whether the given matrix expression is defined.
For those that are defined, give the size of the resulting matrix.
(i) $\operatorname{tr}(DD^T)$

- Based on the provided table, matrix D has dimensions of 4×2.
- Taking the transpose of D reverses these dimensions, resulting in a 2×4 matrix.
- The multiplication of D and its transpose is defined because their inner dimensions match, which results in a 4×4 matrix.

$$D_{4\times2} \times D^T_{2\times4} = (DD^T)_{4\times4}$$

Q. 2(a)

$$\text{Evaluate if } CD^T \text{ is defined and determine its size.}$$

First, we look up the initial dimensions for matrices C and D.

$$C \text{ is } 5 \times 2$$
$$D \text{ is } 4 \times 2$$

Taking the transpose of D swaps its rows and columns.

$$D^T \text{ is } 2 \times 4$$

To see if the multiplication is valid, we check if the inner dimensions match.

$$(5 \times 2) \times (2 \times 4)$$

Since the inner numbers (2 and 2) match, the operation is defined. The final matrix takes the outer dimensions.

$$\text{Size of } CD^T = 5 \times 4$$

Q. 3(j)

Find the trace of:

$$(D - 3E)$$

- We are given specific 3×3 matrices for D and E.

- To evaluate the expression, first multiply matrix E by the scalar 3, and then subtract that result from matrix D.

$$D - 3E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - 3 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$D - 3E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 18 & 3 & 9 \\ -3 & 3 & 6 \\ 12 & 3 & 9 \end{bmatrix} = \begin{bmatrix} -17 & 2 & -7 \\ 2 & -3 & -5 \\ -9 & -1 & -5 \end{bmatrix}$$

  The trace of a matrix is calculated by taking the sum of the entries located on the main diagonal

$$tr(D - 3E) = -17 + (-3) + (-5) = -25$$

Q. 4(a)

$$\text{Evaluate if } 2A^T + C \text{ is defined.}$$

Identify the dimensions of matrix A and flip them to find the dimensions of its transpose, AT. Scalar multiplication does not alter these dimensions.

$$A \text{ is } 4 \times 5 \implies A^T \text{ is } 5 \times 4$$

Matrix C has its given dimensions.

$$C \text{ is } 5 \times 2$$

Set up the addition operation. Matrix addition is only defined when both matrices have the exact same dimensions.

$$(5 \times 4) + (5 \times 2)$$

Because the dimensions do not match, the operation cannot be performed.

$$\text{Undefined}$$

Q. 5(a)

$$\text{Compute } AB$$

We start with the specific matrices given for this section.

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}$$

To compute the product, multiply the rows of A by the columns of B.

$$AB = \begin{bmatrix} (3)(4)+(0)(0) & (3)(-1)+(0)(2) \\ (-1)(4)+(2)(0) & (-1)(-1)+(2)(2) \\ (1)(4)+(1)(0) & (1)(-1)+(1)(2) \end{bmatrix}$$

Simplify the arithmetic inside each element to get your final 3×2 matrix.

$$AB = \begin{bmatrix} 12 & -3 \\ -4 & 5 \\ 4 & 1 \end{bmatrix}$$

Q. 6(f)

In Exercises 3-6, use the following matrices to compute the indicated expression if it is defined.
[Matrices A, B, C, D, E are the same as defined in Q3]
6(f) $D^T E^T - (ED)^T$

- This question asks you to evaluate a specific expression involving transposed matrices.
- A standard property of matrix transposes states that the transpose of a product is equal to the product of the individual transposes in reverse order.

  $$(ED)^T = D^T E^T$$

  If you substitute this property back into the original expression, you are subtracting a matrix from itself, which yields a zero matrix.

  $$D^T E^T - D^T E^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q. 7(c)

In Exercises 7-8, use the following matrices and either the row method or the column method, as appropriate, to find the indicated row or column.

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

7(c) the second column of AB

- The goal is to isolate and find only the second column of the product of matrices A and B.
- Using the column method, you achieve this by multiplying the entirety of matrix A by just the second column vector of matrix B.

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}, \quad B_{col2} = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$$

$$(AB)_{col2} = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$$

$$(AB)_{col2} = \begin{bmatrix} (3)(-2) + (-2)(1) + (7)(7) \\ (6)(-2) + (5)(1) + (4)(7) \\ (0)(-2) + (4)(1) + (9)(7) \end{bmatrix} = \begin{bmatrix} 41 \\ 21 \\ 67 \end{bmatrix}$$

$$(AB)_{col2} = \begin{bmatrix} -6 + (-2) + 49 \\ -12 + 5 + 28 \\ 0 + 4 + 63 \end{bmatrix} = \begin{bmatrix} 41 \\ 21 \\ 67 \end{bmatrix}$$

Q. 8(a)

$$\text{Find the first column of } AB$$

Using the column method, we extract matrix A and only the first column of matrix B.

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}, \quad B_{col1} = \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix}$$

Multiply the entire matrix A by this single column vector.

$$A \times B_{col1} = \begin{bmatrix} (3)(6) + (-2)(0) + (7)(7) \\ (6)(6) + (5)(0) + (4)(7) \\ (0)(6) + (4)(0) + (9)(7) \end{bmatrix}$$

Complete the addition to isolate the first column of the product matrix.

$$A \times B_{col1} = \begin{bmatrix} 18 + 0 + 49 \\ 36 + 0 + 28 \\ 0 + 0 + 63 \end{bmatrix} = \begin{bmatrix} 67 \\ 64 \\ 63 \end{bmatrix}$$

Q. 11(b)

In each part of Exercises 11-12, find matrices A, x, and b that express
the given linear system as a single matrix equation $Ax = b$ and write out this matrix equation.

$$\begin{aligned} 4x_1 \quad & - 3x_3 + x_4 = 1 \\ 5x_1 + x_2 \quad & - 8x_4 = 3 \\ 2x_1 - 5x_2 + 9x_3 - x_4 &= 0 \\ 3x_2 - x_3 + 7x_4 &= 2 \end{aligned}$$

- This linear system consists of equations with four variables.
- To convert this to a matrix equation, map the coefficients into a matrix, placing a 0 wherever a variable is missing from a specific equation.

$$\begin{bmatrix} 4 & 0 & -3 & 1 \\ 5 & 1 & 0 & -8 \\ 2 & -5 & 9 & -1 \\ 0 & 3 & -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

Q. 12(b)

Express the linear system as a single matrix equation $Ax = b$: $5x_1 + 3x_2 + 3x_3 = -5 - x_1 - 3x_2 - 2x_3$

We separate the equations into a coefficient matrix (A), a variable vector (x), and a constants vector (b).

$$A = \begin{bmatrix} 5 & 3 & 3 \\ -1 & -3 & -2 \\ 0 & -4 & 1 \end{bmatrix}$$

Notice that the third equation lacks an x1 term, so its coefficient is 0. We combine everything into the Ax=b format.

$$\begin{bmatrix} 5 & 3 & 3 \\ -1 & -3 & -2 \\ 0 & -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}$$

Q. 13(a)

Express the matrix equation as a system of linear equations:

$$\begin{bmatrix} 5 & 6 & -7 \\ -1 & -2 & 3 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

We perform the matrix multiplication on the left side. Each row of the 3×3 matrix multiplies by the variable vector.

$$\begin{gathered} 5x_1 + 6x_2 - 7x_3 \\ -x_1 - 2x_2 + 3x_3 \\ 0x_1 + 4x_2 - x_3 \end{gathered}$$

Set each resulting expression equal to its corresponding row in the constants vector on the right side.

$$\begin{aligned} 5x_1 + 6x_2 - 7x_3 &= 2 \\ -x_1 - 2x_2 + 3x_3 &= 0 \\ 4x_2 - x_3 &= 3 \end{aligned}$$

**Q. 14(b)

In each part of Exercises 13-14, express the matrix equation as a system of linear equations.

$$\begin{bmatrix} 3 & -2 & 0 & 1 \\ 5 & 0 & 2 & -2 \\ 3 & 1 & 4 & 7 \\ -2 & 5 & 1 & 6 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Here, you are reversing the process by converting a matrix equation back into a system of linear equations.

$$\begin{bmatrix} 3 & -2 & 0 & 1 \\ 5 & 0 & 2 & -2 \\ 3 & 1 & 4 & 7 \\ -2 & 5 & 1 & 6 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Multiply each row of the 4×4 coefficient matrix by the column vector containing the variables.

$$\begin{aligned} 3w - 2x + z &= 0 \\ 5w + 2y - 2z &= 0 \\ 3w + x + 4y + 7z &= 0 \\ -2w + 5x + y + 6z &= 0 \end{aligned}$$

Q. 15

Find all values of k, if any, that satisfy the equation:

$$\begin{bmatrix} k & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

First, multiply the initial 1×3 row vector by the 3×3 matrix.

$$\begin{bmatrix} (k)(1) + (1)(1) + (1)(0) & (k)(1) + (1)(0) + (1)(2) & (k)(0) + (1)(2) + (1)(-3) \end{bmatrix}$$

Simplify this to get a new row vector.

$$= \begin{bmatrix} k+1 & k+2 & -1 \end{bmatrix}$$

Substitute this new vector back into the equation and multiply it by the final column vector.

$$\begin{bmatrix} k+1 & k+2 & -1 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

Expand the dot product into an algebraic equation.

$$(k+1)(k) + (k+2)(1) + (-1)(1) = 0$$

Distribute the terms and combine them to form a standard quadratic equation.

$$k^2 + k + k + 2 - 1 = 0 \implies k^2 + 2k + 1 = 0$$

Factor the perfect square trinomial to solve for k.

$$(k+1)^2 = 0 \implies k = -1$$

Q. 16

$$\begin{bmatrix} 2 & 2 & k \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ k \end{bmatrix} = 0$$

Multiplying the row vector by the 3×3 matrix

$$\begin{bmatrix} 2 & 2 & k \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} (2)(1) + (2)(2) + (k)(0) & (2)(2) + (2)(0) + (k)(3) & (2)(0) + (2)(3) + (k)(1) \end{bmatrix}$$

Simplifying the resulting row vector

$$\begin{bmatrix} 2 + 4 + 0 & 4 + 0 + 3k & 0 + 6 + k \end{bmatrix} = \begin{bmatrix} 6 & 4 + 3k & 6 + k \end{bmatrix}$$

Multiplying the new row vector by the column vector

$$\begin{bmatrix} 6 & 4 + 3k & 6 + k \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ k \end{bmatrix} = 0$$

Expanding the dot product

$$(6)(2) + (4 + 3k)(2) + (6 + k)(k) = 0$$

Simplifying into a standard quadratic equation

$$12 + 8 + 6k + 6k + k^2 = 0$$

$$k^2 + 12k + 20 = 0$$

Factoring the quadratic equation

$$(k + 10)(k + 2) = 0$$

Solving for k

$$k + 10 = 0 \implies k = -10$$

$$k + 2 = 0 \implies k = -2$$